Taha Ahmad Mirza

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Research Interests

Health AI, brain tumour MRI analysis and segmentation, multimodal learning for clinical prediction, physics-informed machine learning, patient-specific computational biomechanics (FEA/CFD), and rigorous evaluation of clinical ML models.

Education

National University of Sciences and Technology (NUST), SINES — Islamabad, Pakistan
Master of Science in Computational Science and Engineering — *Jan 2025 – Jan 2027 (expected)*

- **Thesis:** Patient-Specific Estimation of Tumour-Induced Intracranial Pressure from MRI using Physics-Informed Neural Networks. **Supervisor:** Dr. Zartasha Mustansar.
- **Coursework:** Machine Learning, Deep Learning, Large Language Models, Health AI, Data Analysis and Statistics.

FAST National University of Computer and Emerging Sciences (NUCES) — Islamabad, Pakistan
Bachelor of Science in Computer Science — *Graduated June 2022*

- **Coursework:** Data Structures and Algorithms, Artificial Intelligence, Calculus, Probability and Statistics.

Research Experience

Computational Biomechanics Lab, NUST — Islamabad, Pakistan
Research Scientist (PI: Dr. Zartasha Mustansar) — *Jan 2026 – Present*

- **Building an end-to-end intracranial pressure (ICP) pipeline**: MRI $\rightarrow$ tumour and ventricle segmentation $\rightarrow$ patient-specific 3D brain mesh $\rightarrow$ FEA/CFD simulation $\rightarrow$ neural-network surrogate. ICP is normally measured invasively; the aim is a non-invasive estimate from routine MRI.
- Generated **physics-based synthetic training data**: **9 tumour-growth** $\times$ **CSF-obstruction scenarios per patient** ($\sim$15k mesh nodes) in $\sim$**40 s per case on a laptop**, built as a batch pipeline for **30+ MRI scans**.
- **Competed in the BraTS 2025 glioma segmentation challenge**, training a 3D multi-class model on multi-parametric MRI (T1n, T1c, T2w, T2-FLAIR).

AgriTech Project, SINES, NUST — Islamabad, Pakistan
Research Assistant (PI: Dr. Absaar ul Jabbar) — *Jan 2025 – Dec 2025*

- Built an **Urdu-first crop advisory system** for cotton farmers (Flask + GPT-4.1 **RAG backend**, Android app, Urdu text-to-speech) that turns live soil-sensor readings and weather forecasts into field-specific fertilizer and irrigation advice; **patent filing in progress**.
- Ran a **189-query, 4-arm ablation**: stage-specific rule thresholds reached **87.8%** fully correct advice vs. **74.6%** for full RAG and **5.3%** for the LLM alone; traced the $-$**13.2 pp** drop to retrieval overriding the thresholds (exact McNemar test, $p = 8 \times 10^{-4}$).
- Designed rule-based soil classification (growth-stage NPK limits, **USDA salinity classes** for EC) and 5-day forecast rules for rain and heat stress, served via Flask REST APIs with Firestore sync at $<$ **90 s** end-to-end response.

Publications

Patient-Specific Finite Element and CSF Flow Simulation of Tumour-Induced Intracranial Pressure from MRI — 2026
T. A. Mirza, *Z. Mustansar, et al.* — *Manuscript in preparation*

Selected Projects

Multimodal Alzheimer's Progression Prediction | *Video Vision Transformers, TensorFlow* — Code

- Built a **multimodal pipeline over 487 longitudinal 3D T1 MRI scans (227 subjects)** from ADNI, linking each scan to the clinical visit nearest in time (median gap 5 days), plus demographic and APOE/TOMM40 genetic data.
- **Audited and corrected an earlier pipeline** that reported 81% accuracy: found **subject-level data leakage** (487 scans inflated to 1,212 samples), test-set reuse for early stopping, and label circularity from cognitive-score inputs. Rebuilt it with **subject-level splits, baselines and bootstrap confidence intervals**.

Stable Diffusion Web App | *TensorFlow, Streamlit* Code

- Built a Streamlit text-to-image app and evaluated **generated image quality and prompt alignment** with Hugging Face models and automated metrics.

Data Collection and Training GPT-2 | *GPT-2, BeautifulSoup* Code

- Reduced training loss by 35% via pretraining GPT-2 on 500K+ custom dataset lines from .gov.pk sources.
- Built and fine-tuned tokenizer from scratch to process 3M+ tokens for NLP tasks.

HateSpeechInspector | *Python, XLNet, React, Twint, PyTorch* Code

- Trained an XLNet classifier on a 101K augmented dataset and built a web app that fetches Twitter (X) data in real time and predicts hate categories.

Industry Experience

Afiniti Islamabad, Pakistan
Software Engineer I *Nov 2022 – Jan 2025*

- Analyzed **call data (4.7M+ subscribers)** using SQL, Python, and Excel to identify root causes, **enabling $30K+ monthly revenue recovery**.
- Built a Python tool to automate analysis of call logs, call statistics, sequence diagrams, and error detection.
- Performed unit, feature, UAT, and integration testing of Afiniti products **before AI-based model deployment**.

Open AIMP Remote
Intern *Sept 2021 – Dec 2021*

- Implemented Spark-based data streaming for analytics pipelines, cutting reporting delays from hours to minutes.

Teaching

Computational Biomechanics Lab Islamabad, Pakistan
Graduate TA *Jul 2026 – Aug 2026*

- Facilitated a 6-week internship program on applying AI in Motion Capture and Rapid Prototyping.
- Facilitated an intensive 9-day StemX event introducing AI to O and A level students.

Preply Islamabad, Pakistan
AI Tutor *Nov 2025 – Present*

- Delivered 300+ hours of live instruction across 300+ individual lessons to a global student base of 61 students.
- Holds a 4.6/5 average student rating, a 100% response rate within 24 hours and a 0% lesson cancellation rate.

Honors and Certifications

Research Impact Team Award, Dense Fusion (NUST SINES spin-off), selected from 20+ teams 2025
AWS Certified Cloud Practitioner Credential

Technical Skills

Languages: Python, Bash, SQL
ML and Data: PyTorch, TensorFlow, Hugging Face Transformers, RAG, OpenAI API
Medical Imaging: multi-parametric MRI, NIfTI, nnU-Net, 3D Slicer
Research Methods: medical image segmentation, FEA/CFD simulation, PINNs, ablation studies, significance testing
Tools: Docker, Flask, Firebase/Firestore, Git

References

Dr. Zartasha Mustansar
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